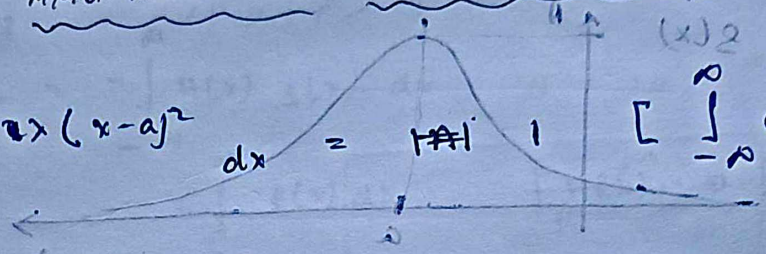
Assignment - 1



② (i) $A \int_{-\infty}^{\infty} e^{-\lambda(x-a)^2} dx = 1$ $\quad \left[\int_{-\infty}^{\infty} \rho(x)\, dx = 1\right]$

$\Rightarrow A\sqrt{\frac{\pi}{\lambda}} = 1.$

$\Rightarrow \boxed{A\sqrt{\frac{\lambda}{\pi}}}$

(ii) $\langle x \rangle = A\int_{-\infty}^{\infty} x\, e^{-\lambda(x-a)^2} dx.$

$\Rightarrow$ Let $y = x - a \Rightarrow dy = dx.$

$\therefore \langle x \rangle = A\int_{-\infty}^{\infty} (y+a)\, e^{-\lambda y^2} dy$

$= A\left[\int y e^{-\lambda y^2} dy + a\int e^{-\lambda y^2} dy\right]$ $\quad \to 0$

$\sqrt{\frac{\lambda}{\pi}} \cdot a \cdot \sqrt{\frac{\pi}{\lambda}} = a$ $\quad \boxed{\langle x \rangle = a}$

$\therefore \langle x^2 \rangle = A\int_{-\infty}^{\infty} x^2 e^{-\lambda(x-a)^2} dx.$

Let $y = x - a \Rightarrow dy = dx.$

$= A\left[\int y^2 e^{-\lambda y^2} dy + 2a\int y e^{-\lambda y^2} dy + a^2\int e^{-\lambda y^2} dy\right]$ $\quad \to 0$

$\therefore$ ~~$\int y^2 e^{-\lambda y^2} dy$~~ $I = \int_{-\infty}^{\infty} e^{-\lambda y^2} dy = \sqrt{\frac{\pi}{\lambda}}$

$$\Rightarrow \frac{dI}{d\lambda} = + \int_{-\infty}^{\infty} y^2 e^{-\lambda y^2} dy = + \frac{1}{2}\sqrt{\pi}\,\lambda^{-3/2} \quad \text{--- (1)}$$

Using (1):

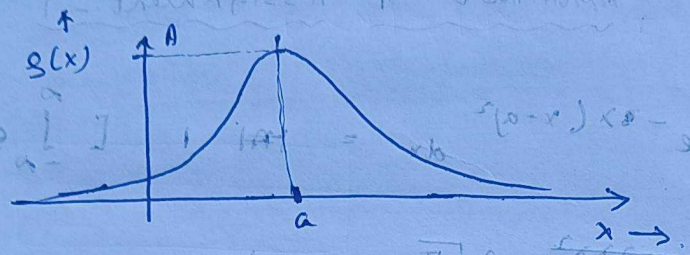
$$\langle x^2 \rangle = A\left[\frac{1}{2}\sqrt{\pi}\,\lambda^{-3/2} + a^2\sqrt{\frac{\pi}{\lambda}}\right].$$

$$= \frac{1}{2}\cdot\sqrt{\pi}\,\lambda^{-3/2}\cdot\frac{\lambda^{1/2}}{\sqrt{\pi}} + a^2 = a^2 + \frac{1}{2\lambda}.$$

$$\boxed{\langle x^2 \rangle = a^2 + \frac{1}{2\lambda}}$$

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \frac{1}{\sqrt{2\lambda}}$$

(iii)



(3) (i) $\int_{-\infty}^{\infty} \delta(x-a) f(x)\, dx = f(a)$ — definition

Let $I_1 = \int_{-\infty}^{\infty} \delta(cx) f(x)\, dx$

Let $u = cx \Rightarrow du = |c|\, dx$.

$\Rightarrow dx = \frac{du}{|c|}$.

$|c|$ is used to preserve orientation.

$u \to \infty$ as $x \to \infty$, but if c is $-ve$

$u \to -\infty$ as $x \to \infty$.

Hence, to preserve the orientation, $|c|$ is used.

$$\therefore I_1 = \frac{1}{|c|}\int_{-\infty}^{\infty} \delta(u) f\left(\frac{u}{c}\right) \frac{du}{|c|} = \frac{f(0)}{|c|} \quad [\text{from def}^n] \quad \text{--- (i)}$$

Also:

$$\frac{1}{|c|}\int_{-\infty}^{\infty} \delta(x) f(x)\, dx = \frac{f(0)}{|c|} = I_2 \quad \text{--- (ii)}$$

~~(i) = (ii)~~ $\Rightarrow I_1 = I_2$

$$\Rightarrow \frac{1}{|c|}\int_{-\infty}^{\infty} \delta(x) f(x)\, dx = \int_{-\infty}^{\infty} \delta(cx) f(x)\, dx.$$

$\Rightarrow$ $\boxed{\delta(cx) = \frac{1}{|c|}\delta(x)}$ [$\because$ $f(x)$ & limits are same, the distribut^n must be the same]

(Proved)

(ii) Let $\theta(x)$ be $H(x)$ [heaviside step f^n].

From integration by parts:

$$\int_{-\infty}^{\infty} f'(x)\, g(x)\, dx = [g(x) f(x)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} g'(x)\, f(x)\, dx$$

$$= -\int_{-\infty}^{\infty} f(x)\, g'(x)\, dx.$$

[assuming the boundary terms vanish]

This is also true for $f(x) = H(x)$. & is written like:

$$\langle H'(x), g(x)\rangle = -\langle H(x), g'(x)\rangle$$

Now, $-\langle H(x), g'(x)\rangle = -\int_{-\infty}^{\infty} H(x)\, g(x)\, dx$

$$= -\int_{0}^{\infty} g(x)\, dx \qquad [H(x) = 0 \;;\; x < 0]$$

$$= -[g(\infty) - g(0)] = g(0)$$

$$\therefore \int_{-\infty}^{\infty} H'(x)\, g(x)\, dx = g(0).$$

This is also the definition of $\delta(x)$. [$\int_{-\infty}^{\infty} \delta(x) g(x)\, dx = g(0)$]

$\boxed{H'(x) = g(x)}$ (Proved)

① (a) $\frac{d|\psi\rangle}{dt} = \frac{1}{i\hbar}\hat{H}|\psi\rangle \Rightarrow \frac{d}{dt}\langle\psi| = -\frac{1}{i\hbar}\langle\psi|\hat{H}$

[$\hat{H}^{\dagger} = \hat{H}$]

$$\cancel{\therefore \frac{d}{dt}\langle\hat{\Omega}\rangle_{\psi} = \left(\frac{d}{dt}\hat{\Omega}\right)|\psi\rangle + \hat{\Omega}\frac{d|\psi\rangle}{dt}}$$

$$\cancel{= \frac{\partial\hat{\Omega}}{\partial t}|\psi\rangle + \frac{\hat{\Omega}\hat{H}}{i\hbar}|\psi\rangle}$$

$$\therefore \frac{d}{dt}\langle\hat{\Omega}\rangle_{|\psi\rangle} = \frac{d}{dt}\langle\psi|\hat{\Omega}|\psi\rangle.$$

$$= \left(\frac{d}{dt}\langle\psi|\right)\hat{\Omega}|\psi\rangle + \langle\psi|\frac{d\hat{\Omega}}{dt}|\psi\rangle + \langle\psi|\hat{\Omega}\left(\frac{d|\psi\rangle}{dt}\right)$$

$$= -\frac{1}{i\hbar}\langle\psi|\hat{H}\hat{\Omega}|\psi\rangle + \left\langle\frac{\partial\hat{\Omega}}{\partial t}\right\rangle + \frac{1}{i\hbar}\langle\psi|\hat{\Omega}\hat{H}|\psi\rangle$$

$$= \left\langle \frac{\partial \hat{\Omega}}{\partial t} \right\rangle + \frac{1}{i\hbar} \langle \psi | [\hat{\Omega}, \hat{H}] | \psi \rangle$$

$$\Rightarrow \frac{d}{dt} \langle \hat{\Omega} \rangle_{|\psi\rangle} = \frac{1}{i\hbar} \langle [\hat{\Omega}, \hat{H}] \rangle_{|\psi\rangle} + \left\langle \frac{\partial \hat{\Omega}}{\partial t} \right\rangle_{|\psi\rangle}$$

(b) If $\hat{\Omega}$ does not explicitly depend on time

$$\left\langle \frac{\partial \hat{\Omega}}{\partial t} \right\rangle = 0$$

If $|\psi\rangle$ is an eigenket of $\hat{H}$; $\hat{H}|\psi\rangle = E|\psi\rangle$.

$$\therefore \langle \psi | [\hat{\Omega}, \hat{H}] | \psi \rangle = \langle \psi | \hat{\Omega}\hat{H} - \hat{H}\hat{\Omega} | \psi \rangle$$

$$= \langle \psi | \hat{\Omega}(\hat{H}|\psi\rangle) - (\langle \psi | \hat{H}) \hat{\Omega} | \psi \rangle$$

$$= E \langle \psi | \hat{\Omega} | \psi \rangle - E^* \langle \psi | \hat{\Omega} | \psi \rangle = 0$$

[$E = E^*$ as H is Hermitian].

$$\boxed{\frac{d}{dt} \langle \hat{\Omega} \rangle_{|\psi\rangle} = 0}$$

(c) (i) $[\hat{\Omega}, \hat{H}] = [\hat{r}\hat{p}, \hat{T} + \hat{V}]$

$$= [rp, T] + [rp, V] \quad \text{— (i)}$$

$$\therefore [rp, T] = \left[rp, \frac{p^2}{2m}\right] = \underbrace{r\left[p, \frac{p^2}{2m}\right]}_{\to 0} + \left[r, \frac{p^2}{2m}\right] p$$

$$= \frac{p}{2m} [r, p^2] \quad \text{— (ii)}$$

$$\therefore [r, p^2] = p[r, p] + [r, p]p$$

$$= 2i\hbar \hat{p}$$

Substituting into (ii).

$$[\hat{\Omega}, \hat{T}] = \frac{p}{2m} \cdot 2i\hbar \hat{p} = 2i\hbar \frac{\hat{p}^2}{2m} = 2i\hbar \hat{T} \quad \text{— (iii)}$$

$$\therefore [\hat{\Omega}, V] = [rp, V]$$

$$= r[p, V] + \underbrace{[r, V]}_{\to 0} p \quad \text{[as } \hat{r} \text{ \& } V(\hat{r}) \text{ should commute]}$$

$$= r[p, V]. \quad \text{— (iv)}$$

$$[\hat{p}, V]\psi = \left[-i\hbar \frac{\partial}{\partial r}(V(r)\psi) + V(r)\, i\hbar \frac{\partial \psi}{\partial r}\right]$$

$$= \left[-i\hbar \frac{\partial V}{\partial r}\psi - \cancel{i\hbar V(r) \frac{\partial \psi}{\partial r}} + \cancel{V(r)\, i\hbar \frac{\partial \psi}{\partial r}}\right]$$

$$\Rightarrow \quad [\hat{p}, v(\hat{r})] = -i\hbar \nabla v(\hat{r}) \frac{\partial}{\partial r}$$

Replacing in (iv):

$$[\hat{\Omega}, v(\hat{r})] = -i\hbar r \nabla v(\hat{r}) \quad \text{— (v)}$$

$$\therefore \quad [\hat{\Omega}, \hat{H}] = 2i\hbar \hat{T} - i\hbar r \nabla v \quad [\text{using (iii) \& (v)}]$$

Now, $\langle \frac{\partial \hat{\Omega}}{\partial t} \rangle = 0$ [no explicit time dependence].

Acting on an energy eigenstate $|\psi\rangle$:

$$\frac{d\langle \hat{\Omega} \rangle}{dt} = 0$$

$$\Rightarrow \quad \frac{1}{i\hbar} \langle [\hat{\Omega}, \hat{H}] \rangle = 0$$

$$\Rightarrow \quad \frac{1}{i\hbar} \langle 2i\hbar \hat{T} - i\hbar r \nabla v \rangle = 0$$

$$\Rightarrow \quad \boxed{2\langle \hat{T} \rangle = \langle \hat{r} \nabla v \rangle} \quad \text{(Proved)}$$

(a) $$v(r) = \frac{-e^2}{4\pi \epsilon r}$$

$$r \nabla v(r) = \frac{e^2}{4\pi \epsilon r} = -v(r)$$

$$2\langle \hat{T} \rangle = \langle r \nabla v \rangle = \langle -v(r) \rangle$$

$$\Rightarrow \quad \boxed{\langle \hat{T} \rangle = -\frac{1}{2} \langle v(r) \rangle} \quad \text{(Proved)}$$